

7.4.33

Harsha-EE25BTECH11026

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Question

A circle C of radius 1 unit is inscribed in an equilateral triangle PQR . The points of contact of C with sides PQ , QR , RP are D , E , F respectively. The line PQ is given by the equation $\sqrt{3}x + y - 6 = 0$ and the point D is $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$. Further, it is given that the origin and the centre of C are on same side of line PQ . The equation of circle C is

- ① $(x - 2\sqrt{3})^2 + (y - 1)^2 = 1$
- ② $(x - 2\sqrt{3})^2 + \left(y + \frac{1}{2}\right)^2 = 1$
- ③ $(x - \sqrt{3})^2 + (y - 1)^2 = 1$
- ④ $(x - \sqrt{3})^2 + (y + 1)^2 = 1$

Theoretical Solution

According to the question,

$$\text{Equation of tangent PQ : } \mathbf{n}^T \mathbf{x} = c \quad (1)$$

$$\text{where } \mathbf{n} = \begin{pmatrix} \sqrt{3} & 1 \end{pmatrix}^T \text{ and } c = 6$$

$$\text{Point of tangency (D) : } \begin{pmatrix} \frac{3\sqrt{3}}{2} \\ \frac{3}{2} \end{pmatrix} \quad (2)$$

$$\text{radius}(r) = 1 \quad (3)$$

Theoretical Solution

From the given data,

$$\mathbf{D} - \mathbf{u} = \lambda \mathbf{n}, \text{ for some scalar } \lambda \quad (4)$$

$$\implies \mathbf{u} = \mathbf{D} - \lambda \mathbf{n} \quad (5)$$

Also,

$$\frac{|\mathbf{n}^T \mathbf{O} - c|}{\|\mathbf{n}\|} = r \quad (6)$$

$$|\mathbf{n}^T \mathbf{u} - c| = r \|\mathbf{n}\| \quad (7)$$

$$\mathbf{n}^T \mathbf{u} = c \pm r \|\mathbf{n}\| \quad (8)$$

Theoretical Solution

To decide the sign, we need to use the fact that the origin and centre of circle are on the same side of the line PQ.

$$\therefore (\mathbf{n}^\top \mathbf{u} - c) \left(\mathbf{n}^\top \begin{pmatrix} 0 \\ 0 \end{pmatrix} - c \right) > 0 \quad (9)$$

$$\implies \mathbf{n}^\top \mathbf{u} < c \quad (10)$$

$$\therefore \mathbf{n}^\top \mathbf{u} = c - r\|\mathbf{n}\| \quad (11)$$

Substituting value of $\mathbf{u}$,

$$\mathbf{n}^\top (\mathbf{D} - \lambda \mathbf{n}) = c - r\|\mathbf{n}\| \quad (12)$$

$$\implies \lambda = \frac{\mathbf{n}^\top \mathbf{D} + r\|\mathbf{n}\| - c}{\mathbf{n}^\top \mathbf{n}} \quad (13)$$

$$\mathbf{u} = \mathbf{D} - \frac{\mathbf{n}^\top \mathbf{D} + r\|\mathbf{n}\| - c}{\mathbf{n}^\top \mathbf{n}} \mathbf{n} \quad (14)$$

Substituting the values,

$$\therefore \mathbf{u} = \begin{pmatrix} \frac{3\sqrt{3}}{2} \\ \frac{3}{2} \end{pmatrix} - \frac{1}{2} \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} = \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} \quad (15)$$

$$\therefore \text{Required equation of circle : } \|\mathbf{x}\|^2 - 2 \begin{pmatrix} \sqrt{3} & 1 \end{pmatrix} \mathbf{x} + 3 = 0 \quad (16)$$

C Code -Finding the equation of circle

```
#include <stdio.h>
#include <math.h>

void circle_equation(double a, double b, double c, double xd,
    double yd, double r) {
    double norm = sqrt(a*a + b*b);
    double nx = a / norm;
    double ny = b / norm;

    double c1x = xd + r * nx;
    double c1y = yd + r * ny;
    double c2x = xd - r * nx;
    double c2y = yd - r * ny;

    double origin_side = a*0 + b*0 + c;
    double c1_side = a*c1x + b*c1y + c;
    double c2_side = a*c2x + b*c2y + c;
    double h, k;
```

C Code -Finding the equation of circle

```
if ((origin_side >= 0 && c1_side >= 0) || (origin_side <= 0
    && c1_side <= 0)) {
    h = c1x; k = c1y;
} else {
    h = c2x; k = c2y;
}

// f = h^2 + k^2 - r^2
double f = h*h + k*k - r*r;

printf("Equation of circle: (x - %.2lf)^2 + (y - %.2lf)^2 =
    %.2lf^2\n", h, k, r);
}
```

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt
import matplotlib as mp
mp.use("TkAgg")
# Load C library
lib = ctypes.CDLL("./circle_equation.so")
# Define argument types
lib.circle_equation.argtypes = [ctypes.c_double, ctypes.c_double,
ctypes.c_double, ctypes.c_double, ctypes.c_double, ctypes.c_double
]
# Define return type (void)
lib.circle_equation.restype = None

a, b, c = np.sqrt(3), 1, -6
xd, yd = 3*np.sqrt(3)/2, 3/2
r = 1
lib.circle_equation(a, b, c, xd, yd, r)
```

```
#plotting
h, k = np.sqrt(3), 1
theta = np.linspace(0, 2*np.pi, 500)
x_circle = h + r*np.cos(theta)
y_circle = k + r*np.sin(theta)
# Line PQ
x_vals = np.linspace(0, 6, 200)
y_vals = -(a*x_vals + c)/b
# Tangency point D
D = np.array([xd, yd])
plt.figure(figsize=(6,6))
plt.plot(x_circle, y_circle, label="Circle C")
plt.plot(x_vals, y_vals, label="Line PQ")
plt.scatter(*D, color="red", label="Tangency D")
plt.scatter(h, k, color="black", label="Center O")
```

```
plt.axis("equal")
plt.grid(True, linestyle="--", alpha=0.6)
plt.legend()
plt.title("Circle Plot")
plt.savefig("/home/user/Matrix/Matgeo_assignments/7.4.33/figs/
Figure_1")
plt.show()
```

Python code

```
import numpy as np
import matplotlib.pyplot as plt
import sympy as sp
import matplotlib as mp
mp.use("TkAgg")

x, y = sp.symbols('x y', real=True)
sqrt3 = sp.sqrt(3)

O = sp.Matrix([sqrt3, 1])
r = sp.Integer(1)
X=np.array([x,y])

expr = (X.dot(X) - 2*O.dot(X) + (O.dot(O) - r**2)).expand()
sp.pprint(expr)
print("=0")
```

Python code

```
h = float(sqrt3.evalf())
k = 1.0
r_num = 1.0

# --- plotting ---
theta = np.linspace(0, 2*np.pi, 500)
x_circle = h + r_num * np.cos(theta)
y_circle = k + r_num * np.sin(theta)

a, b, c = np.sqrt(3), 1.0, -6.0

x_vals = np.linspace(h - 3, h + 6, 400)
y_vals = -(a*x_vals + c) / b

D = np.array([3*np.sqrt(3)/2, 3/2])
plt.figure(figsize=(7,7))
```

Python code

```
plt.plot(x_circle, y_circle, label="Circle C")
plt.plot(x_vals, y_vals, label=r"Line PQ:  $\sqrt{3}x + y - 6 = 0$ ")
plt.scatter(D[0], D[1], color='red', zorder=5, label="D (tangency)")
plt.scatter([h], [k], color='black', zorder=5, label=r"Center  $O(\sqrt{3}, 1)$ ")
plt.gca().set_aspect('equal', 'box')
plt.grid(True, linestyle='--', alpha=0.6)
plt.legend()
plt.title("Equation of Circle C")
plt.xlabel("x")
plt.ylabel("y")
plt.savefig("/home/user/Matrix/Matgeo_assignments/7.4.33/figs/
Figure_1.png")
plt.show()
```

